Shiksha Mandal's
Bajaj Institute of Technology, Wardha
Department of Computer Engineering

Session: 2025-26 **Sem: VI-A**

CA-1 Activity-1: Construction of a Decision Tree

Instructions:

- Construct a decision tree for the assigned dataset.
- Solve the numerical on the A2 page and write it in the form of a poster.
- It must be hand-drawn.

Deadline: Wednesday, 18th March, 2026

Rough Layout of the Poster:

Shiksha Mandal's
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CA-1 Activity-1: Construction of a Decision Tree

Subject: Machine Learning (Even 2025-26)

Roll No: A301 **Student Name: XYZ** **Date of Submission: 18/03/2026**

Dataset:

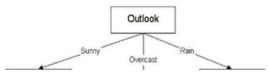
Instance	a1	a2	a3	Class
1	True	Hot	High	No
2	True	Hot	High	No
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No
7	True	Hot	High	No
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes

...

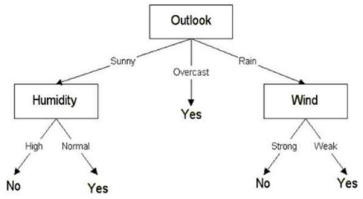
Iteration N:

Formulas:

Partial Decision Tree with a root:



Final Decision Tree:



Iteration 1: To find root of the tree

Iteration 2:

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Que 1 for Roll Nos A301-309

For a SunBurn dataset given below, construct a decision tree

Name	Hair	Height	Weight	Location	Class
Sunita	Blonde	Average	Light	No	Yes
Anita	Blonde	Tall	Average	Yes	No
Kavita	Brown	Short	Average	Yes	No
Sushma	Blonde	Short	Average	No	Yes
Xavier	Red	Average	Heavy	No	Yes
Balaji	Brown	Tall	Heavy	No	No
Ramesh	Brown	Average	Heavy	No	No
Swetha	Blonde	Short	Light	Yes	No

Que 2 for Roll Nos A310-318

With the following data set, generate a decision tree and predict the class label for a data point with values <Female, 2, standard, high>.

Gender	Car Ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Que 3 for Roll Nos A319-327

Write and explain decision tree for the following transactions:

Tid	Refund	Martial status	Taxable Income	Cheat
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	Yes	Divorced	220 K	No
8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes